

Asphaltenes: Predicting and Preventing Wellbore Plugging

Flow assurance in production wellbores — and the seed of my graduation project

Presenter: [Your full name] · ID [xxxxxxx]

Supervisor: [Prof. name]

Programme: BSc Petroleum and Gas Engineering —
[University]

[Month] 2026 · 15-minute presentation · 23 slides +
2 backup

ONE-LINE SUMMARY

Field engineers call asphaltenes the cholesterol of crude oil. This seminar shows when they block wells, how we predict it — and the project that will quantify it for a real case.

Asphaltenes are a billion-dollar flow-assurance problem

billions \$/yr

industry-wide cost of deposition —
Farooq et al., Energy & Fuels (2021)
[1]

≈ \$70 M

one deepwater GoM well: shut-in +
cleanup — Farooq et al. (2021) [1]

**up to \$1.2
M/day**

deferred production during a
plugging event — Stratiev et al.,
Processes (2025) [2]

up to $\frac{2}{3}$

of tubing radius: measured
deposits, Hassi Messaoud —
SPE-994-PA (1965) [3]

When production pressure drops, the heaviest molecules in the crude crash out of solution and build a hard, tar-like deposit that chokes the tubing.

Not an exotic failure mode: asphaltenes are managed somewhere in the world every day — North Sea, Gulf of Mexico, Algerian Sahara.

The sting is not only the money — the deposit arrives unplanned, and the industry's default answers are reactive.

So the question of this seminar: can we predict WHERE it will deposit — before the well chokes?

What are asphaltenes? The heaviest, most polar fraction of crude — held in suspension

■ Operational definition (not one molecule, but a solubility class):

insoluble in n-heptane, soluble in toluene — the basis of the SARA analysis (Saturates · Aromatics · Resins · Asphaltenes): ASTM D6560 / IP 143 [12].

■ Why they stay in the oil:

aromatic-rich resins form the stabilising shell around nano-aggregates of ~2 nm; the crude is stable while that dispersion is maintained (Yen-Mullins picture) [11].

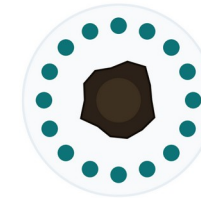
■ Why the nickname fits:

field engineers call them the cholesterol of crude oil — and like cholesterol, the danger is not their presence, it is precipitation from circulation.

■ Scale:

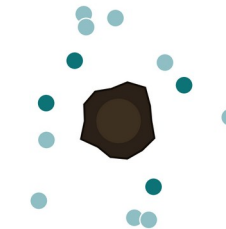
typically a few per cent of the oil by weight — enough to plug a well completely.

1 · STABLE — the resin / aromatic shell is intact



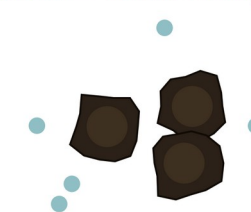
The shell keeps the asphaltene cores dispersed in the oil.

2 · DESTABILISED — the shell is stripped away



Pressure drop or injected gas removes the protective shell.

3 · FLOCCULATED — cores clump and can stick

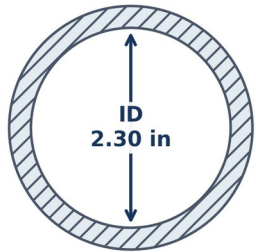


Precipitation is not deposition — but flocculation enables it.

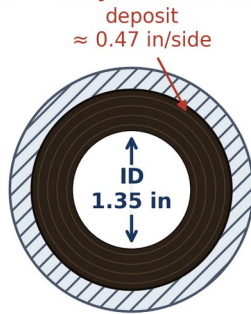
Inside the tubing: two-thirds of the flow area, gone

Tubing damage, three views — two-thirds of the bore can disappear

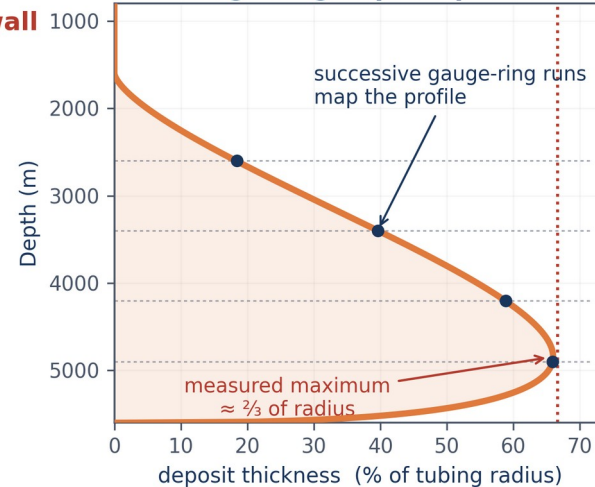
BEFORE — clean bore



AFTER — deposit on the wall



Gauge-ring deposit profile



Cross-sections to scale from the measured ID reduction 2.30 → 1.35 in (Wylde & Punase, JPT, 2020).

Gauge-ring profile: schematic after the Hassi Messaoud field practice (Haskett & Tartera, SPE-994-PA, 1965).

Left and centre: cross-sections drawn TO SCALE from the measured ID reduction 2.30 → 1.35 in [4].

Right: gauge-ring deposit profile — the field method that maps where the deposit sits, panel by panel, so treatment volumes can be calculated [3].

If you can obtain a real cut-tubing photograph with permission, it replaces the cross-sections — keep the profile panel either way.

■ Flow area shrinks.

ID 2.30 → 1.35 in: about two-thirds of the flow area lost [4].

■ Pressure is consumed downhole.

Severe fields lost 20–25 % of wellhead pressure within 15–20 days [3].

■ Production declines, then stops.

A choking well can cease flowing within days.

■ Invisible until expensive.

No downhole window — the pressure signature arrives late.

Objective: explain the mechanism, review the defenses, propose the project

1 · EXPLAIN

what destabilises asphaltenes,
and the chain from precipitation
to deposition

2 · REVIEW

today's defenses — and why they
are mostly reactive

3 · PROPOSE

the graduation project: predicting
onset depth and injection point for
a real well

Scope statement: this is a flow-assurance review and a feasibility study for the thesis — not a completed field intervention study.

Saying this out loud protects you in the Q&A: the committee knows exactly what you claim and what you do not.

How the next 14 minutes are organised

PART 1 The problem: what asphaltenes are, and what they cost 0:00 - 2:45

PART 2 The science: stability, triggers, and the P-T envelope 2:45 - 6:20

PART 3 Field evidence: a documented deepwater case study 6:20 - 9:20

PART 4 Defenses then and now — and the economics of prevention 9:20 - 11:20

PART 5 The graduation project: prediction workflow and plan 11:20 - 14:25

SCOPE — a flow-assurance review plus a thesis feasibility study; not a completed field intervention.

Stability is a fragile balance — pressure, temperature and composition all pull on it

■ Two engineering views, one working rule:

colloidal (aggregates protected by resins) and solubility (asphaltenes dissolve best in aromatic, dense fluids).

■ Field screening uses solubility parameters:

condensed into the Colloidal Instability Index [5] (Yen et al.; threshold scale after Asomaning) — higher CII, less stable crude.

■ Three levers move the balance:

pressure, temperature, and composition — gas or condensate changes the oil's solvent quality.

■ Engineering consequence:

anything that lightens the oil or drops its pressure pushes it toward instability.

THE THREE LEVERS

PRESSURE ↓

main trigger — loss of light-end solvency as P falls

COMPOSITION changed

gas lift, gas breakthrough, CO₂ / rich-gas injection, commingling

TEMPERATURE

secondary modifier — shifts the envelope, direction is oil-specific

$$\text{CII} = (\text{Saturates} + \text{Asphaltenes}) / (\text{Resins} + \text{Aromatics})$$

orange numerator = the destabilising fraction · teal denominator = the crude's own dispersant · higher CII = less stable crude

Asphaltenes are not wax — the trigger, the physics and the fix all differ

	Paraffin wax	Asphaltenes
Main trigger	temperature falls below the wax appearance temperature (WAT)	pressure falls below the onset pressure; composition change
Nature of the solid	crystalline — it can melt	amorphous — it does not remelt; thermal methods largely fail
Where solids appear	in flowlines and surface lines, cold spots	wherever the flowing P-T path crosses the envelope — often deep in the tubing
Deposition mechanism	diffusion + shear; hard wax layer grows inward	adhesion of flocculated aggregates; can form throughout the flow path
Removal	hot oil, thermal insulation, scrapers	aromatic solvents (toluene/xylene), milling, dispersants
Inhibition	crystal modifiers / pour-point depressants	polymeric dispersants (inhibitors) injected continuously below the onset

NEXT → how that instability becomes a deposit in the tubing.

Temperature matters — but as a modifier of the envelope, not the primary trigger. Juries test this distinction.

From stable oil to plugged tubing: four steps — and the critical distinction



1 · PRECIPITATION

Thermodynamic. The oil crosses the envelope and asphaltenes separate from solution as nano-aggregates.



2 · FLOCCULATION

Particles stick together into larger, irregular clusters.



3 · AGGREGATION

Clusters grow; some can approach millimetre scale.



4 · DEPOSITION

Particles adhere to the steel and form a deposit — balanced by shear removal.

PRECIPITATION ≠ DEPOSITION

Many wells carry precipitated asphaltenes to surface with no deposit at all. Deposition is a transport phenomenon on top of the thermodynamic one — governed by adhesion, particle size and shear. This is exactly where prediction is weakest today.

The envelope: precipitation begins the moment the flowing path crosses the AOP

■ AOP:

asphaltene onset pressure — above the line the oil is stable.

■ Above the bubble point:

the upper onset sits above P_b and precipitation peaks near P_b — instability windows reach thousands of psi [6, 15].

■ Below P_b :

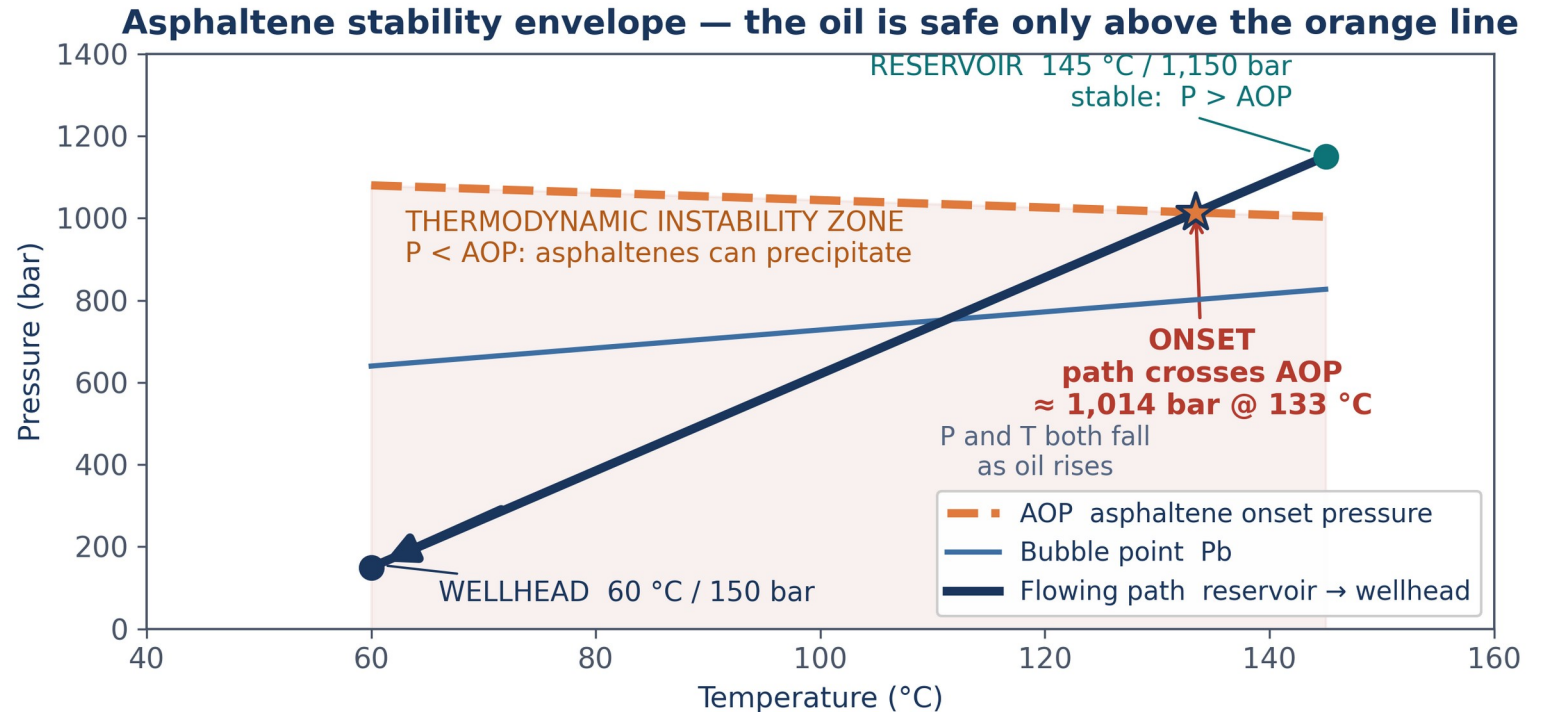
asphaltenes partly re-dissolve; some oils then show a second, lower onset (backup slide).

■ Unstable region:

below the AOP line, precipitation is thermodynamically possible.

■ The flowing path:

as oil rises, P and T fall together — the crossing is the onset.



Illustrative envelope reconstructed from the published deepwater Gulf of Mexico case data (Wylde & Punase, JPT, 30 Apr 2020); P-T path illustrative.

How risk is measured: screening first, onset measurement second

Method	What it gives you	Practical role
SARA analysis (ASTM D2007-class)	the four fractions; feeds CII and solubility modelling	first-pass screen on any new crude [12]
CII / stability indices	a single number: stable vs unstable tendency	flags the crude before any lab work [5]
De Boer plot	AOP vs bubble point vs reservoir pressure	field-level screening; if AOP > Pb, expect problems [6]
Automated heptane titration (ASTM D6703)	the asphaltene precipitation onset in titrations	quantitative ranking of crudes and blends [12]
High-pressure onset measurement (NIR / light scattering in a PVT cell)	the AOP as a function of temperature — the input a simulator must match	the calibration anchor for every model [4]

The best field practice uses several: a screening number for early decisions, and a measured AOP curve to calibrate the prediction. My project uses both.

What triggers it in the wellbore — and where deposits actually form

■ Pressure drop — the main trigger.

As oil rises, pressure falls; when the path crosses the AOP, precipitation begins.

■ Gas lift changes the solvent.

Injected gas strips resins off the aggregates; the oil loses solvency.

■ Gas breakthrough / CO₂ / rich gas

shifts the envelope — CO₂ is a documented destabiliser.

■ Commingling and blending

individually stable crudes can form an unstable mixture.

■ Acid jobs and workovers

spent acids destabilise the near-well region.

CORRECTING A CLASSIC MISCONCEPTION

“Deposits form at the top of the well, where the pressure drops fastest.” Not true — the pressure gradient along the tubing is roughly steady, so the deposit forms wherever the flowing path first crosses the envelope. In deep, hot wells that is often TENS of metres — sometimes a kilometre — below the wellhead.

Why it matters: that depth decides where you inject inhibitor — and the field case study (next slide) shows what happens when the injection point is too shallow.

NEXT → the field case that shows the cost of getting that depth wrong.

GoM case study: injection placement above the onset left 3,400 ft unprotected

■ The well (all depths labelled on the figure):

deepwater GoM; perfs 17,700–18,000 ft; mandrel 14,600 ft [4].

■ The problem:

3,400 ft of tubing BELOW the mandrel had no chemical protection.

■ The model:

effective tubing ID fell from 2.3 in to 1.35 in while producing.

■ The fix:

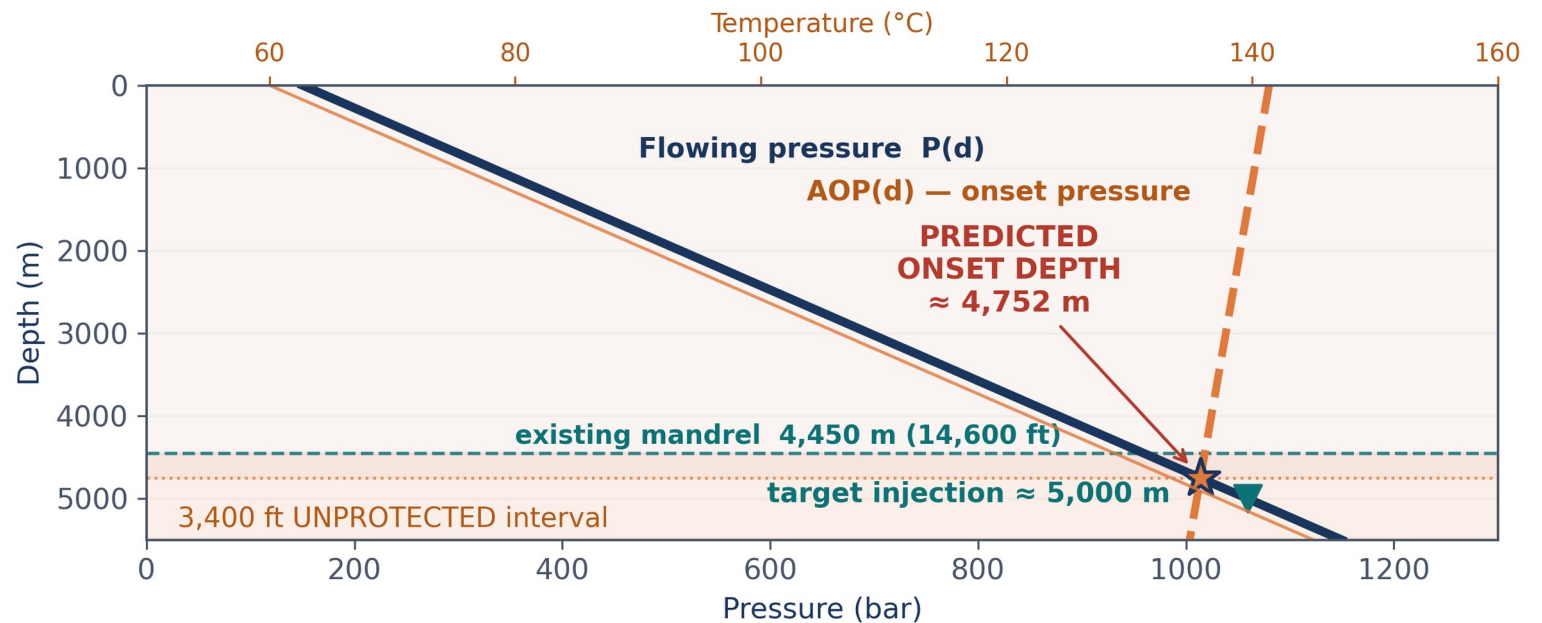
solvent soak, then continuous inhibitor — 750 ppm, stabilised near 500 ppm.

■ The result:

+16 % production over 4 months, including shut-in time [4].

NEXT → what you can actually do about it.

Where it deposits — the number that sets the injection point



Illustrative P-T model. Geometry — perforations 17,700–18,000 ft, mandrel 14,600 ft — from the documented deepwater Gulf of Mexico case (Wylde & Punase, JPT, 30 Apr 2020).

Reconstructed from published data: geometry (perforations 17,700–18,000 ft; mandrel 14,600 ft) from Wylde & Punase, JPT (2020) [4]. The P-T model itself is illustrative — the thesis rebuilds it with the field's own PVT data.

Detecting the problem before the well chokes

■ Rate and pressure:

a rising ΔP for the same rate ($\Delta P/Q$) is the earliest fingerprint of a growing deposit.

■ Wellhead signature:

severe fields lose 20–25 % of wellhead pressure in 15–20 days — a clean, quantitative alarm [3].

■ Wireline gauge rings / calipers:

successive runs map the deposit profile and volume, so treatment volumes can be sized [3].

■ Sampling:

optical analysis of produced fluids tuned the inhibitor dose in near-real time in the case well [4].

■ Monitoring + periodic CT cleanouts:

deliberate surveillance where prevention alone is not trusted [4].

Reactive cleanouts are temporary, expensive — and guarantee repeat NPT

Method	How it works	Why it is not the answer
Wireline scrapers / cutting	mechanical removal of the deposit from tubing	slow, impractical in bad wells; routine scraping was 'expensive and cumbersome' at Hassi Messaoud [3]
Coiled-tubing milling / jetting	mechanical cleanout of the full string	rig-up + NPT; the well is shut in while it happens
Aromatic solvent washes (toluene / xylene)	dissolves the deposit; volume calculated from gauge-ring profile	HSE exposure (BTEX), flammability, cost; only a temporary fix — deposits return
Acid / stimulation jobs	dissolve scale around the near-wellbore	can destabilise the crude and worsen asphaltene damage — used with great care

Thermal methods are not a melting story — asphaltene deposits have no melting point, and heat can even weaken the resins that keep them dispersed. That is the wax toolkit, not this one [15].

Every reactive method is a cleanout — and every cleanout is NPT, deferred production and a repeat visit scheduled by the physics itself.

The proactive answer: keep the asphaltenes dispersed — don't let them land

■ Chemical inhibitors:

polymer/surfactant molecules adsorb onto the aggregates — polar head on the particle, alkyl tail into the oil — and keep them suspended. Inhibitors shift the flocculation onset; dispersants shrink the aggregates. Both prevent formation; neither dissolves an existing deposit [15, 16].

■ Delivery — capillary injection string:

thin tubing clamped to the production string, dosing continuously at depth with a surface pump.

■ Dosage is field-specific:

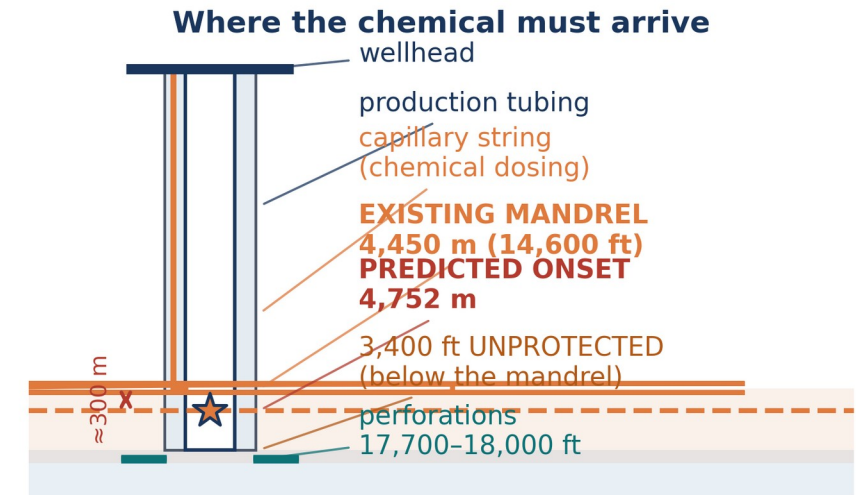
hundreds to a few thousand ppm; here 750 ppm initial, stabilised $\approx$ 500 ppm [4].

■ Placement rule (the lesson of Part 3):

the injection point must sit **BELOW** the deepest credible onset depth — onset plus a margin sized by the sensitivity study (backup slide 24). Rules of thumb leave tubing unprotected.

■ Chemical-free alternative:

re-injecting dead oil restores solvent quality — promising, still developing [7].



Schematic, not to scale. Case geometry after Wylde & Punase, JPT, 2020. Onset depth from the illustrative model on slide 13.

WHY IT PAYS

No rig-up, no deferred production, no repeat workover. The well keeps producing while the problem is managed continuously — and the schematic shows exactly where the chemical has to arrive.

The business case: reactive cleanouts vs continuous inhibition

Cost element	Reactive approach	Proactive approach
Routine cost	treatment documented up to $\approx$ \$3 M/well — Stratiev et al. (2025) [2]	inhibitor: \$31–46 k/well/yr (Middle East) to \$330–390 k/well/yr (GoM) — Cenegy (2001) survey, via Khaleel et al. (2020) [8, 9]
Deferred production	up to $\approx$ \$1.2 M/day during a plugging/shut-in event — Stratiev et al. (2025) [2]	near zero — the well keeps producing
Catastrophic tail	$\approx$ \$70 M/well shut-in + cleanup (GoM); up to \$100 M if the deposit forms at the SCSSV — Farooq et al. (2021); Elsevier monograph (2021) [1, 10]	residual risk only if dosage/placement is wrong
Frequency	scheduled by the physics — deposits return	managed continuously; occasional CT verification runs



The thesis supplies the missing breakeven frequency for this specific well — the cleanout rate at which continuous inhibition pays for itself. Today that number is argued by rule of thumb.

Attribution: the inhibitor ranges trace to Cenegy's 2001 survey; Fluid Phase Equilibria (2020) and the Elsevier monograph (2021) assign the high range to the GoM, Energy & Fuels (2021) to the Middle East — I quote the majority and flag the discrepancy.

Gap analysis: prediction is still the weak link

- Onset prediction is sensitive to fluid characterisation — small input errors move the AOP and therefore the predicted depth.
- Precipitation $\neq$ deposition: transport/adhesion models are far less mature than the thermodynamics.
- Inhibitor dosage is still chosen largely by trial and error — published criticism: lab validation methods are inadequate, and field results can disappoint [7].
- Injection-point placement is rarely optimised against a predicted onset depth — the case study showed 3,400 ft left unprotected [4].
- No consolidated, public workflow connects screening → onset curve → deposition depth → injection point → dosage → economics.

THE OPPORTUNITY

A transparent, auditable workflow — every assumption visible, every number reproducible — applied end-to-end on a documented well. At Bachelor's level, documented rigour is the contribution.

Graduation project: onset and deposition-depth prediction for a real well

AIM: build and document a transparent screening → PVT → wellbore workflow that predicts asphaltene onset depth, selects the inhibitor injection point, and quantifies the economics — validated against a documented field case.

RESEARCH QUESTIONS

RQ1 How sensitive is predicted onset depth to fluid characterisation (SARA/CII inputs, EoS choice, AOP data)?

RQ2 What injection point and dosage window keep the whole tubing above the AOP — with what uncertainty margin?

RQ3 At what cleanout frequency does continuous inhibition break even for the case well?

WORK PACKAGES

WP1 Fluid screening: SARA, CII, de Boer assessment of the case fluid

WP2 EoS modelling: asphaltene onset curve (CPA / PC-SAFT class model), tuned to measured AOP

WP3 Wellbore model: P-T profile → onset depth → injection-point design

WP4 Economics: breakeven frequency and reactive-vs-proactive comparison

Methodology: screening → modelling → validation → economics

1 · SCREEN

- SARA + CII + de Boer on published fluid data
- Output: stability verdict and risk flag

2 · MODEL

- Asphaltene onset curve vs temperature (CPA/PC-SAFT-class EoS)
- Tune to any published/measured AOP
- Output: the envelope

3 · PREDICT

- Wellbore P-T profile → crossing depth
- Sensitivity: $\pm\Delta P$, $\pm AOP$, $\pm GOR$
- Output: onset depth + margin

4 · DECIDE

- Injection point + dosage window
- Breakeven cleanout-frequency model
- Output: decision matrix for the field

1 · LICENSED, if available PVTsim ·
Multiflash · OLGA / PIPESIM

2 · OPEN-SOURCE FALLBACK NeqSim
(CPA-EoS) + Python — verified

3 · FLOOR correlation-based screening

TOOL HIERARCHY — the project cannot stall on software: licence check is a Week-1 action, and route 2 is already verified.

Deliverables, honest limitations, and a realistic 16-week plan

DELIVERABLES

- Thesis with full calculation chain
- Documented prediction workflow (spreadsheet / Python)
- Case-study report: onset depth and injection point
- Breakeven economics model
- Decision matrix: monitor / inhibit / clean out

LIMITATIONS & ETHICS (stated up-front)

- Literature-based fluids — no live sample unless a partner provides one
- Single documented case: conditional conclusions
- Deposition-transport modelling kept first-order
- Steady-state wellbore profiles
- Cost figures indicative, from published ranges
- Solvent chemistries compared on paper only — no uncontrolled BTEX handling, no vendor recommended

Weeks	Milestone
1 - 3	Literature, standards, case selection
4 - 6	Screen fluid; build the envelope model
7 - 9	Wellbore model; onset-depth prediction
10 - 11	Validation against the case; sensitivity
12 - 14	Economics + thesis writing
15 - 16	Review, defence preparation

NEXT → three messages to take away.

Three messages — then your questions

1

Asphaltene plugging is a pressure-and-composition story, not a cooling story — and precipitation is not the same as deposition.

2

The decisive number is the depth where the flowing path crosses the onset envelope: it sets the injection point, the mitigation strategy and the money.

3

My graduation project builds a transparent screening-to-economics workflow for that depth — validated on a documented field case.

Thank you — I am happy to take your questions.

Key references (full list per course citation style)

- [1] Farooq, U.; Lædre, S.; Gawel, K. Review of Asphaltenes in an Electric Field. *Energy & Fuels* 2021, 35 (9), 7285–7304. doi:10.1021/acs.energyfuels.0c03962
- [2] Stratiev, D.; Nikolova, R.; Veli, A.; Shishkova, I.; Toteva, V.; Georgiev, G. Mitigation of Asphaltene Deposit Formation via Chemical Additives: A Review. *Processes* 2025, 13 (1), 141. doi:10.3390/pr13010141
- [3] Haskett, C. E.; Tartera, M. A Practical Solution to the Problem of Asphaltene Deposits — Hassi Messaoud Field, Algeria. *J. Pet. Technol.* 1965, 17 (4), 387–391. SPE-994-PA. doi:10.2118/994-PA
- [4] Wylde, J.; Punase, A. Asphaltenes: A Complex and Challenging Flow Assurance Issue To Measure and Quantify Risk. *J. Pet. Technol.*, 30 Apr 2020 (deepwater GoM case study).
- [5] Yen, A.; Yin, Y. R.; Asomaning, S. Evaluating Asphaltene Inhibitors: Laboratory Tests and Field Studies. SPE 65376, 2001. Thresholds after Asomaning (2003): CII $\geq$ 0.9 unstable; $<$ 0.7 stable.
- [6] de Boer, R. B.; Leerlooyer, K.; Eigner, M. R. P.; van Bergen, A. R. D. Screening of Crude Oils for Asphalt Precipitation: Theory, Practice and the Selection of Inhibitors. *SPE Prod. Facil.* 1995, 10 (1), 55–61. doi:10.2118/24987-PA
- [7] Khaleel, A. T.; Abutayya, M. I. L.; Sisco, C. J.; Vargas, F. M. Mitigation of Asphaltene Deposition by Re-injection of Dead Oil. *Fluid Phase Equilib.* 2020, 514, 112552. doi:10.1016/j.fluid.2020.112552
- [8] Cenegy, L. M. Survey of Successful World-Wide Asphaltene Inhibitor Treatments in Oil Production Fields. SPE ATCE, New Orleans, 2001 (original source of the inhibitor cost data).
- [9] Asphaltene Deposition Control by Chemical Inhibitors, 1st ed.; Gulf Professional/Elsevier: 2021; Ch. 1 (cost table: removal \$0.3–3.5 M/well; sidetrack $\approx$ \$50 M; downtime $\approx$ \$700 k/day).
- [10] Critical Analysis of Different Techniques Used To Screen Asphaltene Stability in Crude Oils. *Fuel* 2021, 299, 120839. [Author list to confirm at library]
- [11] Mullins, O. C. The Modified Yen Model. *Energy & Fuels* 2010, 24 (4), 2179–2207; reviewed in Mullins et al., *Energy & Fuels* 2012. doi:10.1021/ef300185p
- [12] ASTM D6560-22 / IP 143:21 (asphaltenes, n-heptane insolubles); D2007 (clay-gel SARA); D4124 (four fractions); D3279 (n-heptane insolubles); D6703 (automated Heithaus titrimetry).
- [13] NeqSim — open-source thermodynamic toolkit (CPA-EoS; asphaltene screening). Equinor, Apache-2.0 licence, v3.22. github.com/equinor/neqsim
- [14] Evaluation and Modeling of Asphaltene Deposition in Oil Wells. SPE-206366-MS, SPE ATCE, Dubai, 2021. doi:10.2118/206366-MS
- [15] Mohammed, I.; Mahmoud, M.; Al Shehri, D.; El-Husseini, A.; Alade, O. Asphaltene Precipitation and Deposition: A Critical Review. *J. Pet. Sci. Eng.* 2021, 197, 107956 (thermal/steam limits; inhibitor mechanisms). doi:10.1016/j.petrol.2020.107956
- [16] Kelland, M. A. Production Chemicals for the Oil and Gas Industry. CRC Press, 2009 (inhibitor-versus-dispersant distinction; accessed via AADE-24-FTCE-073).

Full citation details, exact quotes and page references: 04_Source_Log.docx. Technical cross-check, risk register and sensitivity study: 06_Technical_Accuracy_and_Risk_Register.docx. Keyed to every number in the deck.

Derivation and assumptions: envelope, indices and the onset-depth model

■ CII definition:

$CII = (\text{Saturates} + \text{Asphaltenes}) / (\text{Resins} + \text{Aromatics})$; boundary ≈ 0.9 unstable, < 0.7 stable — thresholds are laboratory-specific.

■ De Boer plot (correctly stated):

in-situ density (x) vs gas under-saturation, $P_{\text{res}} - P_{\text{b}}$ (y); three risk zones; risk greatest for LIGHT, undersaturated crudes. The onset-above-Pb rule is a SEPARATE envelope screen.

■ Onset curve and depth model:

AOP(T) is linear here for illustration — the thesis tunes a CPA / PC-SAFT-class EoS to measured AOP. Onset depth = where $P(d)$ falls below AOP(T(d)); steady-state profiles here, multiphase in the thesis.

■ Sensitivity study (quantified):

base 4,752 m. ± 50 bar $\rightarrow \pm 255$ m $\cdot \pm 100$ bar $\rightarrow \pm 511$ m $\cdot \pm 10\%$ gradient $\rightarrow -404 / +487$ m $\cdot$ gas-lift (gradient -10% , AOP $+50$ bar) $\rightarrow 5,520$ m: the whole tubing unstable, so the DOSE does the work. Design rule: inject below the deepest credible onset.

■ Deliberately excluded:

deposition growth (first-order only), lower-AOP phenomena, near-wellbore damage.

Data sources, tools and validation plan

■ Fluid data:

published PVT/SARA reports from case-study literature; field case in Wylde & Punase, JPT (2020); Hassi Messaoud classic data (SPE-994-PA); other published well databases via the university library.

■ Licensed tools (if available):

PVT simulator with asphaltene onset (PVTsim / Multiflash / WinProp-class) and a wellbore flow model (OLGA with asphaltene module / PIPESIM). Confirm with the department in Week 1.

■ Open-source fallback:

NeqSim (Equinor, Apache-2.0) — CPA-EoS asphaltene screening and PVT; Python for the workflow, plots and the economics model.

■ Validation targets:

reproduce the documented case behaviour — predicted onset below the existing mandrel; ID-reduction trend; dosage within the field's 750→500 ppm window — within a stated tolerance; cross-check onset against correlations.

■ Risk register (top 3):

(1) no live fluid sample → use published data and declare it; (2) licensed software unavailable → NeqSim + correlations (project already designed for this); (3) case-data conflicts across sources → document the variance and choose conservative values.

■ Ethics & HSE framing:

aromatic solvents (BTEX) are a workplace-safety topic; the project compares alternatives explicitly and never recommends handling without controls.